Recurrence Formula (I): The Three-Term Relation

- **Claim:** $(2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x)$.
- **Proof:** Differentiate the generating function $(1-2xt+t^2)^{-1/2} = \sum_n P_n t^n$ with respect to t:

$$(x-t)(1-2xt+t^2)^{-3/2} = \sum_n nP_n t^{n-1}.$$

- Multiply both sides by $(1-2xt+t^2)$ and use the generating function again to replace $(1-2xt+t^2)^{-3/2}(1-2xt+t^2)^{-1} \to$ substitute and simplify:

$$(x-t)\sum_n P_n t^n = (1-2xt+t^2)\sum_n nP_n t^{n-1}.$$

- Equating coefficients of t^n on both sides gives the three-term relation.



Recurrence Formula (I): Result and Remark

- Result:

$$\boxed{(2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x)}$$

- This is the classical **three-term (pure) recurrence relation** for Legendre polynomials.
- **Remark:** replacing n by $n-1$ gives the equivalent form

$$nP_n(x) = (2n-1)xP_{n-1}(x) - (n-1)P_{n-2}(x).$$



Recurrence Formula (II): Differential Recurrence

- **Claim:** $xP_n'(x) - P_{n-1}'(x) = nP_n(x)$.
- **Proof:** Differentiate the generating function with nrespect to t and, separately, with respect to x:

$$(x-t)(1-2xt+t^2)^{-3/2} = \sum_n nP_n t^{n-1}, \qquad t(1-2xt+t^2)^{-3/2} = \sum_n P_n' t^n.$$

- Multiply the second relation by $(x-t)/t$ and compare with the first; matching coefficients of t^n gives the identity

$$\boxed{xP_n'(x) - P_{n-1}'(x) = nP_n(x)}$$

- This is a **differential recurrence relation**: it relates P_n' and P_{n-1}' directly, without an intervening integral or sum.



Recurrence Formula (III) and (IV)

- **Formula (III):** Differentiating Formula (I) and combining with Formula (II) (eliminating P'_{n-1}) gives

$$\boxed{(2n+1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)}$$

- **Formula (IV):** Writing (II) and (III) together and subtracting gives

$$\boxed{(n+1)P_n(x) = P'_{n+1}(x) - xP'_n(x)}$$

- These are both **differential** recurrence relations, expressing derivatives of one P_n in terms of neighboring polynomials (not their derivatives).



Recurrence Formula (V) and (VI)

- **Formula (V):** replacing $n \to n-1$ in (IV) and combining with (II) gives

$$\boxed{(1-x^2)P_n'(x) = n\big[P_{n-1}(x) - xP_n(x)\big]}$$

- **Formula (VI):** rewriting (I) and combining with (V) gives the equivalent form

$$\boxed{(1-x^2)P_n'(x) = (n+1)\big[xP_n(x) - P_{n+1}(x)\big]}$$

- These express $(1-x^2)P_n'(x)$ two different ways — useful for computing derivatives of P_n without differentiating the explicit polynomial.



Beltrami's Result

- Combining Formulas (V) and (VI): multiply (V) by $(n+1)$ and (VI) by n, then add:

$$(2n+1)(1-x^2)P_n'(x) = n(n+1)\big[P_{n-1}(x) - P_{n+1}(x)\big].$$

- This is known as **Beltrami's Result**:

$$\boxed{(2n+1)(1-x^2)P_n'(x) = n(n+1)\big[P_{n-1}(x) - P_{n+1}(x)\big]}$$

- It is a convenient combined form of the two previous recurrence relations, often used to evaluate integrals involving P_n'.



Example: Evaluating $\int_{-1}^{1} x^2 P_{n+1} P_{n-1}\, dx$

- Apply the triple recursion with $n \to n-1$ and $n \to n+1$ separately:

$$(2n-1)xP_{n-1} = nP_n + (n-1)P_{n-2},$$

and

$$(2n+3)xP_{n+1} = (n+2)P_{n+2} + (n+1)P_n.$$

- Multiply the two equations together and integrate over $[-1,1]$; every cross term vanishes by orthogonality except the $P_n \cdot P_n$ term:

$$(2n+3)(2n-1)\int_{-1}^{1} x^2 P_{n+1} P_{n-1}\, dx = n(n+1)\int_{-1}^{1} P_n^2\, dx = n(n+1)\frac{2}{2n+1}.$$

- Hence

$$\int_{-1}^{1} x^2 P_{n+1}(x) P_{n-1}(x)\, dx$$

$$= \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}.$$



Example: Evaluating $\int_{-1}^{1} x^2 P_n^2(x)\,dx$

- Square the triple recursion formula and integrate over $[-1, 1]$:

$$(2n+1)^2 \int_{-1}^{1} x^2 P_n^2\,dx = (n+1)^2 \int_{-1}^{1} P_{n+1}^2\,dx + n^2 \int_{-1}^{1} P_{n-1}^2\,dx$$
$$+ 2n(n+1) \int_{-1}^{1} P_{n+1} P_{n-1}\,dx.$$

- The last integral is 0 by orthogonality (indices differ by 2). Using $\int P_k^2\,dx = \frac{2}{2k+1}$:

$$(2n+1)^2 \int_{-1}^{1} x^2 P_n^2\,dx = (n+1)^2 \frac{2}{2n+3} + n^2 \frac{2}{2n-1}.$$

- Hence

$$\int_{-1}^{1} x^2 P_n^2(x)\,dx = \frac{2(n+1)^2}{(2n+3)(2n+1)^2}$$
$$+ \frac{2n^2}{(2n-1)(2n+1)^2}.$$



Problem : Evaluating $\int_{-1}^{1} xP_n(x)P_{n-1}(x)\,dx$

- Use the triple recursion formula $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$ to write
$$xP_n(x) = \frac{(n+1)P_{n+1}(x) + nP_{n-1}(x)}{2n+1}.$$
- Multiply by $P_{n-1}(x)$ and integrate over $[-1,1]$:
$$\int_{-1}^{1} xP_nP_{n-1}\,dx = \frac{n+1}{2n+1}\int_{-1}^{1} P_{n+1}P_{n-1}\,dx + \frac{n}{2n+1}\int_{-1}^{1} P_{n-1}^2\,dx.$$
- The first integral is 0 by orthogonality; the second is $\frac{2}{2n-1}$. So
$$\int_{-1}^{1} xP_n(x)P_{n-1}(x)\,dx = \frac{n}{2n+1}\cdot\frac{2}{2n-1} = \frac{2n}{(2n+1)(2n-1)}.$$
- Check at $n=1$: $\int_{-1}^{1} xP_1P_0\,dx = \int_{-1}^{1} x^2\,dx = \frac{2}{3}$, and the formula gives $\frac{2}{3\cdot 1} = \frac{2}{3}$. ✓



Problem : Evaluating $\int_{-1}^{1} P_n(x)\,dx$

- Evaluate $\displaystyle\int_{-1}^{1} P_n(x)\,dx$ for all $n \geq 0$.
- For $n = 0$: $\displaystyle\int_{-1}^{1} P_0(x)\,dx = \int_{-1}^{1} 1\,dx = 2.$
- For $n \geq 1$: write the integrand as $P_n(x) \cdot P_0(x)$ (since $P_0(x) \equiv 1$) and apply orthogonality directly:

$$\int_{-1}^{1} P_n(x)\,dx = \int_{-1}^{1} P_n(x)P_0(x)\,dx = 0 \qquad (n \neq 0).$$

- So

$$\int_{-1}^{1} P_n(x)\,dx = \begin{cases} 2 & n = 0, \\ 0 & n \geq 1. \end{cases}$$



Problem : Evaluate $I_n = \int_{-1}^{1} [P_n'(x)]^2 \, dx, \qquad n \geq 1.$

- A useful identity for Legendre polynomials is

$$P_n'(x) = \sum_{\substack{k=0 \\ n-k \text{ odd}}}^{n-1} (2k+1) P_k(x).$$

- Using the orthogonality relation,

$$\int_{-1}^{1} P_k(x) P_j(x) \, dx = \frac{2}{2k+1} \delta_{kj},$$

we obtain

$$I_n = \int_{-1}^{1} \left[P_n'(x)\right]^2 \, dx = \sum_{\substack{k=0 \\ n-k \text{ odd}}}^{n-1} (2k+1)^2 \frac{2}{2k+1}$$

$$= 2 \sum_{\substack{k=0 \\ n-k \text{ odd}}}^{n-1} (2k+1).$$



Solution (cts.)

- Case 1: $n = 2m$ Then $k = 1, 3, \ldots, 2m-1$. Therefore,

$$I_{2m} = 2\sum_{j=1}^{m}(4j-1) = 2\left[4\frac{m(m+1)}{2} - m\right] = 2m(2m+1).$$

- Since $n = 2m$,

$$I_n = n(n+1).$$

- Case 2: $n = 2m+1$ Then $k = 0, 2, \ldots, 2m$. Thus,

$$I_{2m+1} = 2\sum_{j=0}^{m}(4j+1) = 2\left[4\frac{m(m+1)}{2} + (m+1)\right]$$

$$= (2m+1)(2m+2).$$

- Again, $I_n = n(n+1)$.
- Hence,

$$\boxed{\int_{-1}^{1}\left[P_n'(x)\right]^2\,dx = n(n+1), \qquad n \geq 1.}$$



Evaluate $I_n = \int_{-1}^{1} P_1(x)P_n(x)P_{n+1}(x)\,dx$.

- Since
$$P_1(x) = x,$$
we have
$$I_n = \int_{-1}^{1} xP_n(x)P_{n+1}(x)\,dx.$$
- Using the recurrence relation,
$$xP_n(x) = \frac{n+1}{2n+1}P_{n+1}(x) + \frac{n}{2n+1}P_{n-1}(x),$$
we obtain
$$I_n = \frac{n+1}{2n+1}\int_{-1}^{1} P_{n+1}^2(x)\,dx + \frac{n}{2n+1}\int_{-1}^{1} P_{n-1}(x)P_{n+1}(x)\,dx.$$



Solution (ctn)

- The second integral vanishes by orthogonality:

$$\int_{-1}^{1} P_{n-1}(x)P_{n+1}(x)\,dx = 0.$$

- Furthermore,

$$\int_{-1}^{1} P_{n+1}^2(x)\,dx = \frac{2}{2n+3}.$$

- Therefore,

$$\boxed{\int_{-1}^{1} P_1(x)P_n(x)P_{n+1}(x)\,dx = \frac{2(n+1)}{(2n+1)(2n+3)}.}$$



Problem : Proving $P'_{n+1}(x) - P'_{n-1}(x) = (2n+1)P_n(x)$

- This is the companion recurrence used in Christoffel's expansion — let's verify it directly for small n.
- $n = 1$: $P'_2(x) - P'_0(x) = 3x - 0 = 3x = 3P_1(x)$. ✓
- $n = 2$:
  $P'_3(x) - P'_1(x) = \frac{1}{2}(15x^2 - 3) - 1 = \frac{15x^2-5}{2} = 5 \cdot \frac{3x^2-1}{2} = 5P_2(x)$. ✓
- $n = 3$: $P'_4(x) - P'_2(x) = \frac{1}{8}(140x^3 - 60x) - 3x =$
  $\frac{35x^3-15x}{2} - 3x + 3x = 7 \cdot \frac{5x^3-3x}{2} = 7P_3(x)$. ✓
- In each case, the identity $P'_{n+1}(x) - P'_{n-1}(x) = (2n+1)P_n(x)$ holds — this is exactly the recurrence used (in reindexed form) in Christoffel's expansion.



Christoffel's Expansion: Statement

- **Christoffel's Expansion:**

$$P_n'(x) = (2n-1)P_{n-1}(x)+(2n-5)P_{n-3}(x)+(2n-9)P_{n-5}(x)+\cdots,$$

where the last term is $3P_1(x)$ if n is even, or $P_0(x)$ if n is odd.

- This expresses the derivative $P_n'(x)$ as a finite sum of *undifferentiated* Legendre polynomials — very useful for term-by-term integration.



Christoffel's Expansion: Proof

- From recurrence Formula (III), $P'_{n+1} = (2n+1)P_n + P'_{n-1}$. Replacing $n \to n-1$:

$$P'_n = (2n-1)P_{n-1} + P'_{n-2}. \qquad \text{(A)}$$

- Replacing n by $n-2, n-4, \ldots, 4, 2$ successively in (A) gives a telescoping chain:

$$P'_{n-2} = (2n-5)P_{n-3} + P'_{n-4}, \quad \ldots, \quad P'_2 = 3P_1 + P'_0 = 3P_1 \text{ (since } P'_0 = 0).$$

- Adding all these equations telescopes the $P'_{n-2}, P'_{n-4}, \ldots$ terms away, leaving (for n even)

$$P'_n = (2n-1)P_{n-1} + (2n-5)P_{n-3} + \cdots + 3P_1.$$

- For n **odd**, the chain ends instead at $P'_3 = 5P_2 + P'_1 = 5P_2 + P_0$ (since $P'_1 = 1 = P_0$), giving the stated $P_0(x)$ ending term.

